

Temperature Matters: Thermal Feedback for Awe Experiences in VR

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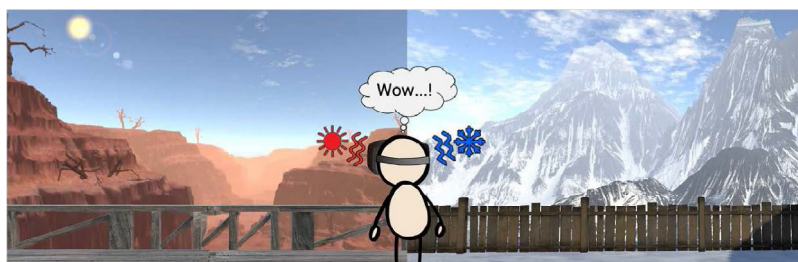


Figure 1: Using vast natural environments and matching temperature feedback to enhance the experience of awe in virtual reality: feeling the warmth of a desert canyon (left) and the chill of a snow-covered mountain (right).

ABSTRACT

This work explores how thermal feedback can enhance awe experiences in virtual reality (VR). We developed a custom thermal feedback system integrated into a VR headset that delivers temperature sensations to the user's face while viewing vast scenes of snow-covered mountains and desert canyons. Our results show that thermal feedback significantly enhanced presence measures while influencing specific components of awe experiences, specifically those related to physical sensations.

Index Terms: Virtual reality, awe, emotion, thermal feedback.

1 INTRODUCTION

Virtual reality (VR) has emerged as a powerful tool for inducing complex emotions like awe through immersive environments. Awe is characterized by feelings of vastness and a need for cognitive accommodation when encountering extraordinary experiences [8]. These awe experiences often manifest through physical sensations such as goosebumps and self-reported chills [5]. Research has shown that VR can effectively induce awe by enhancing presence, with higher levels of presence leading to more intense awe experiences [1]. Given that awe experiences are closely tied to bodily responses like chills, thermal feedback could provide an additional sensory channel for enhancing these physical aspects of awe. Although thermal feedback has been shown to enhance presence in virtual environments [6], its potential for contributing to awe experiences remains unexplored.

To investigate this potential, we developed a thermal feedback system for the Meta Quest 2 Head Mounted Display (HMD) using Peltier elements to deliver temperature changes to the user's face. We evaluated our approach by creating two distinct awe-inspiring natural environments with matching thermal feedback and investigating their effects on subjective experiences of awe and presence.

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2 RELATED WORK

2.1 Awe Elicitation in VR

VR can transport users to otherwise inaccessible experiences, making it particularly effective for inducing awe [3]. Studies have demonstrated that VR environments elicit stronger awe responses compared to traditional media presentations [1]. Particularly effective are virtual natural landscapes [5], with mountain environments showing strong potential for evoking awe through their ability to convey vastness [2].

2.2 Head-Mounted Thermal Feedback in VR

Research has explored thermal feedback integration in VR using Peltier elements. Previous work used five thermal modules mounted on the HMD's facial interface for directional temperature cues [4]. Later designs refined this approach using six Peltier elements to allow for more precise temperature control [6]. Alternative methods using electrical stimulation of the forehead have also been explored, though with limited success in achieving consistent temperature perception [7].

3 PILOT STUDY

We conducted a 2x2 within-subjects experiment with scene (snow mountain, canyon) and thermal stimuli (on, off) as variables. Participants experienced these four conditions in randomized order using a balanced Latin square design: snow mountain with and without cold stimuli, and canyon with and without heat stimuli.

3.1 Thermal feedback system

We designed a custom 3D-printed facial interface for the Meta Quest 2 that integrates five Peltier elements (see Figure 2 for the setup). Following previous work [4], three elements (40 x 20 x 3.4mm), are positioned on the forehead, with two additional elements (30 x 15 x 4.2mm), placed below each eye. The thermal system is controlled via an Arduino UNO through an L298N H-bridge module. Temperature control uses pulsed cycles: 4 seconds on and 6 seconds off for hot sensations, 5 seconds on and off for cold sensations. Without active cooling mechanisms, these intervals prevent the Peltier elements from continuous operation and overheating [6].



Figure 2: Peltier elements mounted on the HMD facial interface. Forehead elements (P1-P3) and eye-level elements (P4-P6).

3.2 Virtual Environments

The experience consists of two mountain-based scenes designed to elicit awe through environmental vastness [2]. The canyon scene (Figure 1, left) depicts an open desert canyon in sun-heated terrain, while the snow mountain scene (Figure 1, right) presents a vast winter landscape with snow-covered peaks. Both scenes include environmental effects that support temperature perception: wind and snowfall in the mountain scene, and heat distortion and sun glare in the canyon. The thermal feedback is synchronized with the scene exploration, adding temperature stimuli to the audiovisual experience. The environments were implemented in Unity 2021.3.6f1 for Meta Quest 2 (90Hz refresh rate, 1832 x 1920 pixels per eye).

3.3 Measures

At the beginning of the experiment, participants provided demographic information (age, gender, and VR experience). Awe was assessed after each condition using the Situational Awe Scale (SAS), which measures four dimensions of awe: Connection, Oppression, Chills, and Diminished Self. Presence was measured using the IGroup Presence Questionnaire (IPQ), which assesses presence across four subscales: General Presence, Spatial Presence, Involvement, and Realism. Both questionnaires used 7-point scales.

3.4 Procedure

Each condition started with a 20-second adaptation period (black screen) to help participants mentally prepare, followed by 60 seconds of scene exploration with matching thermal stimuli. Participants removed the HMD between conditions to complete the questionnaires, preventing thermal carry-over effects between stimuli.

4 RESULTS

Eight volunteers (2 females, 6 males, aged 22-39, $M = 29.0$, $SD = 4.69$) participated in the pilot study. We compared the responses between thermal and non-thermal conditions across both environments to examine how thermal feedback affects presence and awe.

Analysis of the SAS data (Figure 3, left) comparing thermal and non-thermal conditions revealed a significant difference in the *Chills* subscale ($W = 16.5$, $p < .05$, $r_{rb} = 0.438$). The *Connection* subscale showed a trend toward significance ($W = 31.5$, $p = .065$, $r_{rb} = 0.5$). No significant differences were found for *Oppression* ($W = 24.5$, $p = .758$, $r_{rb} = 0.0$) or *Diminished Self* ($W = 43.0$, $p = .551$, $r_{rb} = 0.125$). Analysis of the IPQ data (Figure 3, right) compared thermal and non-thermal conditions for presence using a Wilcoxon Signed-Rank test, revealing significant differences in *General Presence* ($W = 0.0$, $p < .05$, $r_{rb} = 0.75$), *Spatial Presence* ($W = 3.0$, $p < .05$, $r_{rb} = 0.75$), and *Realism* ($W = 13.5$, $p < .05$, $r_{rb} = 0.5$). No significant difference was found for the *Involvement* subscale ($W = 46.5$, $p = .71$, $r_{rb} = 0.125$).

5 DISCUSSION AND FUTURE WORK

Our results show that thermal feedback can influence both presence and awe in VR. The significant increase in the *Chills* subscale of the SAS suggests a relationship between thermal feedback

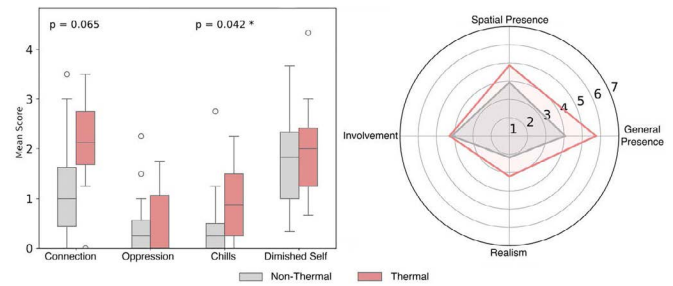


Figure 3: SAS awe ratings (left) and IPQ presence ratings (right) between thermal and non-thermal conditions.

and the emotional response of awe, extending previous work on VR-induced awe [2] by exploring how additional sensory feedback might contribute to the experience. However, the distinction between physical temperature sensations and emotional chills (warm, cold, or moving) requires careful interpretation. The *Connection* subscale showed a trend towards significance, indicating a possible relationship between thermal feedback and users' connection to virtual environments, though further investigation with larger sample sizes is needed. The presence results further support the value of thermal feedback, with significant improvements in *General Presence*, *Spatial Presence*, and *Realism*. These findings align with previous work on thermal feedback in VR [6]. Our extension to awe-inducing environments also aligns with earlier findings suggesting that enhanced presence facilitates the elicitation of awe [1].

Future work should include physiological measurements like skin conductance and heart rate variability to understand awe responses better [1]. Further research could explore adaptive thermal patterns and integration with other sensory feedback to create more intense awe experiences in VR.

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REFERENCES

- [1] A. Chirico, P. Cipresso, D. B. Yaden, F. Biassoni, G. Riva, and A. Gaggioli. Effectiveness of immersive videos in inducing awe: an experimental study. *Scientific reports*, 7(1):1218, 2017. 1, 2
- [2] A. Chirico, F. Ferrise, L. Cordella, and A. Gaggioli. Designing awe in virtual reality: An experimental study. *Frontiers in psychology*, 8:2351, 2018. 1, 2
- [3] A. Chirico, D. B. Yaden, G. Riva, and A. Gaggioli. The potential of virtual reality for the investigation of awe. *Frontiers in psychology*, 7:1766, 2016. 1
- [4] R. L. Peiris, W. Peng, Z. Chen, L. Chan, and K. Minamizawa. Thermovr: Exploring integrated thermal haptic feedback with head mounted displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 5452–5456, 2017. 1
- [5] D. Quesnel and B. Riecke. Are you awed yet? how virtual reality gives us awe and goose bumps. *Frontiers in psychology*, 9:2158, 2018. 1
- [6] K. Ragozin, X. Meng, R. Lalintha Peiris, K. Wolf, G. Chernyshov, and K. Kunze. Thermoquest-a wearable head mounted display to augment realities with thermal feedback. In *Proceedings of the 20th International Conference on Mobile and Ubiquitous Multimedia*, pp. 62–66, 2021. 1, 2
- [7] T. Saito, J. Zhang, T. Kameoka, and H. Kajimoto. Thermal sensation on forehead using electrical stimulation: Thermal sensation using electrical stimulation. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–5, 2021. 1
- [8] M. N. Shiota, D. Keltner, and A. Mossman. The nature of awe: Elicitors, appraisals, and effects on self-concept. *Cognition and emotion*, 21(5):944–963, 2007. 1